Capstone Design

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Verified Study Buddy

THE CONNECTED WEEK

One connected capstone-design record.

Explain requirements, review the corpus boundary, audit proof claims and specify a test.

Keep one habit: Keep user need, baseline, corpus/version, response rules, owner and expected evidence together.

MAIN LESSON · SECTION 1

Recall: reconnect earlier ideas

> Retrieve before opening notes.

Retrieve and explain

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Human AI Workflow. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Bias and Representation. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Red Team the Product. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

My remembered idea, example and correction

My remembered idea, example and correction

▼ **Earlier wording, alternatives and picture transcripts**

Human AI Workflow

Bias and Representation

Red Team the Product

W1 · Explain a buildable brief

> Connect need, corpus, contract, owner and tests.

Your evidence record

- Design review W2: student room assistant uses public register P and current approval A, excludes teacher-only T; manual same-room approval/capacity lookup baseline. Main glossary V1: root anchors plant, stem supports leaves, leaf captures light for photosynthesis. V2 adds approved flower reproductive-structure entry and retains V1 entries.

  Contract supplies supported entry+reference, clarification for standalone ambiguity, explicit fallback for absent/out-of-scope facts. Plans are not observed results.

- Reference for this exercise: open this week's student lesson, "Capstone Design", at "See how it works" and "Follow a worked example". Use the original worked values or code named in the question; the later pictorial case is a separate example. Keep this reference beside your workbook while practising.

- Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

> **My reasoning and evidence**

MAIN LESSON · SECTION 13

W2 · Review the room-corpus design

> Keep the repeated scenario labelled development review.

Your evidence record

- Design review W2: student room assistant uses public register P and current approval A, excludes teacher-only T; manual same-room approval/capacity lookup baseline. Main glossary V1: root anchors plant, stem supports leaves, leaf captures light for photosynthesis. V2 adds approved flower reproductive-structure entry and retains V1 entries.

  Contract supplies supported entry+reference, clarification for standalone ambiguity, explicit fallback for absent/out-of-scope facts. Plans are not observed results.

- Student room assistant corpus contains public register P, teacher-only guide T and current approval A. It must report approved room capacity or needs_review. Show the intermediate steps and give the conclusion with its scope.

> **My reasoning and evidence**

MAIN LESSON · SECTION 14

W3 · Audit plan-as-proof claims

> Implementation outcomes still need recorded checks.

Your evidence record

- Design review W2: student room assistant uses public register P and current approval A, excludes teacher-only T; manual same-room approval/capacity lookup baseline. Main glossary V1: root anchors plant, stem supports leaves, leaf captures light for photosynthesis. V2 adds approved flower reproductive-structure entry and retains V1 entries.

  Contract supplies supported entry+reference, clarification for standalone ambiguity, explicit fallback for absent/out-of-scope facts. Plans are not observed results.

- A report omits this qualification: A complete capstone plan is not **evidence** that the implementation meets it; evaluate after building.

  Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

> **My reasoning and evidence**

MAIN LESSON · SECTION 16

W5 · Design a revealing requirement test

> Supply evidence versions, expected behavior and missing/ambiguous boundary.

Your evidence record

- Design review W2: student room assistant uses public register P and current approval A, excludes teacher-only T; manual same-room approval/capacity lookup baseline. Main glossary V1: root anchors plant, stem supports leaves, leaf captures light for photosynthesis. V2 adds approved flower reproductive-structure entry and retains V1 entries.

  Contract supplies supported entry+reference, clarification for standalone ambiguity, explicit fallback for absent/out-of-scope facts. Plans are not observed results.

- Reference for this exercise: open this week's student lesson, "Capstone Design", at "See how it works" and "Follow a worked example". Use the original worked values or code named in the question; the later pictorial case is a separate example. Keep this reference beside your workbook while practising.

- For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

  Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

MAIN LESSON · SECTION 4

W4 · Design the new science-note request behavior independently

> Keep one first attempt before hints or feedback.

Read and explain independently

Treat the three quoted requests as separate standalone requests, each with no previous conversation state. Only the two supplied science notes are permitted; label them with source IDs of your choice and keep the references consistent. No weather record, new source or treatment question is supplied.

A science assistant may use only these notes: evaporation is liquid becoming vapour; condensation is vapour becoming liquid. Specify behavior for “Define evaporation”, “Explain it”, and “What is tomorrow’s weather?”.

> **My independent explanation, trace and limits**
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Keep a fair record

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

▼ **Earlier wording, alternatives and picture transcripts**

Independent case: Changed independent case

MAIN LESSON · SECTION 9

Project · Keep the decision and its evidence

> Add this week’s change to your existing project record.

Your evidence record

Freeze the Study Buddy v1 design and acceptance tests before implementation.

The project change and **evidence** I will keep

> **Write or draw your reasoning and evidence.**

MAIN LESSON · SECTION 17

Feedback · Repair and return

> Keep the first attempt, the repair and a changed-case return.

Your evidence record

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

MAIN LESSON · SECTION 10

Finish · Judge the evidence in your work

> Point to work that supports your confidence level.

Your evidence record

- I can explain the mechanism.
- I can show evidence.
- I can handle a changed case.
- I can name a limit.

Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> This is the same W2 attempt. Keep one response record.

Use this when helpful

Use W2: Trace the worked case for Capstone **requirement**s and corpus. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

▼ Extra practice · Repair the explanation

REFERENCE · SECTION 3

Extra practice · Repair the explanation

> Explain the deciding fact before discussing the answer.

Use this when helpful

Use the worked case for Capstone **requirement**s and corpus. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Original learning target

REFERENCE · SECTION 5

Reference · Original learning target

> Use this target to inspect an explanation.

Use this when helpful

Design a chapter assistant with one answerable, one ambiguous and one unanswerable query. State the expected behaviour and one mandatory quality criterion.

▼ **Extra practice · A complete case**

REFERENCE · SECTION 6

Extra practice · A complete case

> Try the question before checking the worked explanation.

Use this when helpful

Capstone: answer from one approved chapter with citations and a missing-**evidence** fallback.

Define acceptance tests.

Show your trace, calculation or **evidence**

Write or draw your reasoning and evidence.

▼ **Extra practice · A changed case**

REFERENCE · SECTION 7

Extra practice · A changed case

> Keep this practice separate from the reserved independent attempt.

Use this when helpful

The capstone goal is helps students. Rewrite it as an observable test for a passage-based study assistant.

My practice answer and reason

> **Write or draw your reasoning and evidence.**

Week 33 Test and reflect

▼ **Extra practice · Design a revealing test**

REFERENCE · SECTION 8

Extra practice · Design a revealing test

> Predict before tracing or testing.

Use this when helpful

Create a changed **input**, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record	My evidence
Changed input or condition	
Expected answer or fallback	
Observed result	
Why the result occurred	

▼ **Reference · Original target response space**

REFERENCE · SECTION 11

Reference · Original target response space

> This teaching case is practice rather than a fresh assessment.

Use this when helpful

Design a chapter assistant with one answerable, one ambiguous and one unanswerable query. State the expected behaviour and one mandatory quality criterion.

My explanation after practice

Record support used: none, hint or teacher explanation.

▼ Additional practice · Earlier equipment-corpus plan

REFERENCE · SECTION 15

Additional practice · Earlier equipment-corpus plan

> This retained task differs from the selected standalone science-note assessment.

Earlier task and response

- Student equipment assistant has public inventory I, restricted purchase ledger L and public booking notice B. Define eligible corpus, baseline and a missing-**evidence** test.

My reasoning and evidence

▼ Extra practice · Use the worked picture

REFERENCE · SECTION 18

Extra practice · Use the worked picture

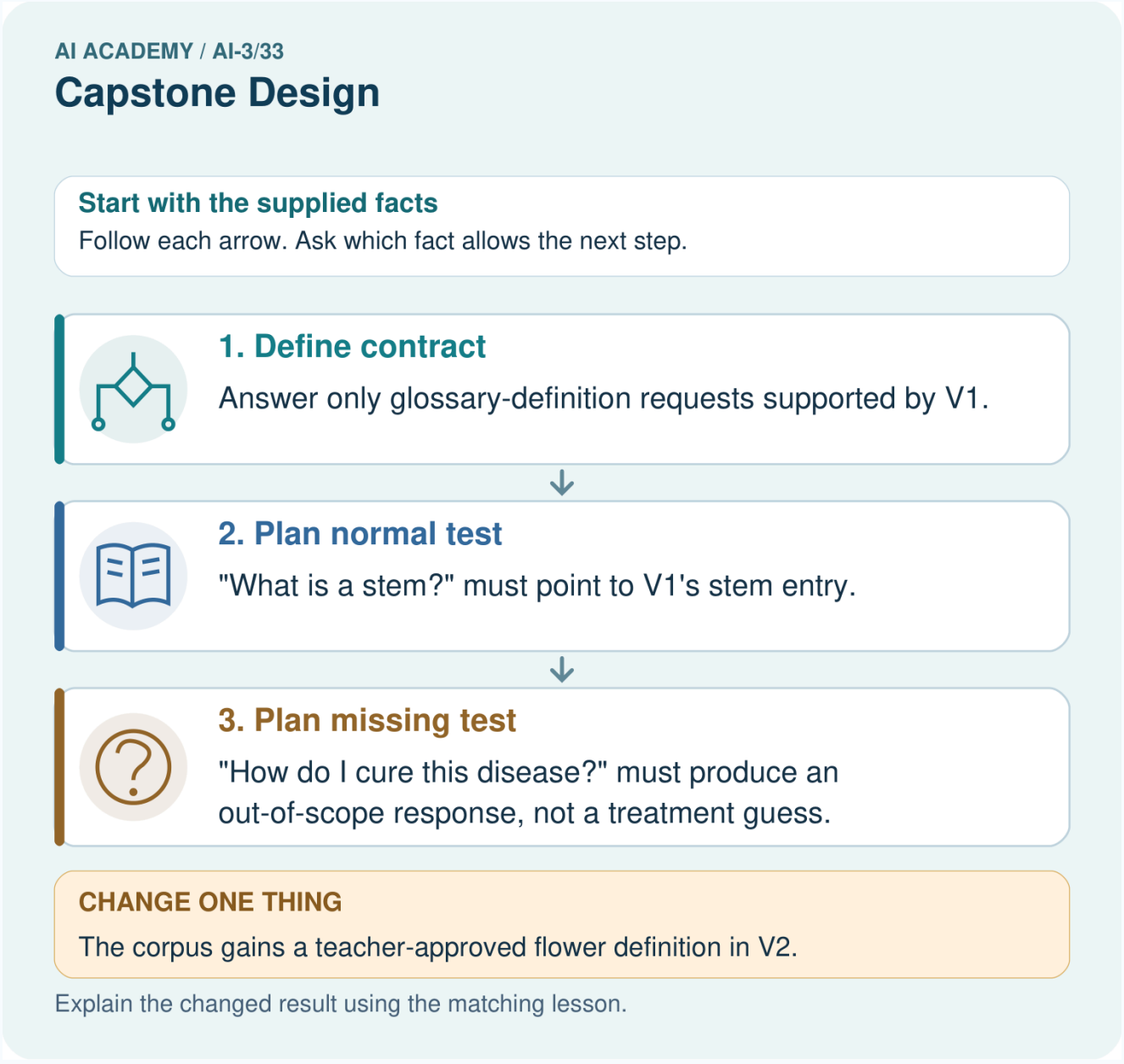
> Keep the explanation and the support you used.

Use this when helpful

Plan a bounded study buddy around a small approved vocabulary corpus.

Corpus V1 defines roots, stem and leaf. It contains no plant disease treatment. Success requires a source reference; missing definitions require a clear fallback.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.



Follow the labelled picture with the matching explanation. Open the vector diagram

What should the project plan record before the new flower test? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

▼ **Reference · Project portfolio record**

REFERENCE · SECTION 19

Reference · Project portfolio record

> Extend your main project response when helpful.

Use this when helpful

Project: Verified Study Buddy

Current milestone: Present and defend. Demonstrate the system, explain one failure and justify the next improvement with **evidence**.

For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

> **Explain what this week adds to your project. Reuse last week's record rather than starting again.**

▼ **Extra practice: pictures, repair and feedback**

REFERENCE · SECTION 20

Extra practice: pictures, repair and feedback

> Attempt each question before opening its explanation. Use one browser or paper copy.

First supported question

- Plan a bounded study buddy around a small approved vocabulary corpus.
- Which labelled step matches this detail? "How do I cure this disease?" must produce an out-of-scope response, not a treatment guess.
- 1. Plan missing test

- 2. Plan normal test
- 3. Define contract
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.
- After your attempt, compare with the supported explanation: Plan missing test “How do I cure this disease?” must produce an out-of-scope response, not a treatment guess. Read the picture in order.

  “How do I cure this disease?” must produce an out-of-scope response, not a treatment guess. The arrows show which recorded **input** or intermediate result each decision uses.

  Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Practice 1: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**
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▼ **Check the first explanation**

Practice 2: explain the deciding evidence

Changed-case question

- Changed-case practice: The corpus gains a teacher-approved flower definition in V2. Which conclusion is supported by the supplied facts?
- 1. This one example guarantees correct results for every future **input**.

- 2. Add a flower test using V2 and record the corpus version.
- 3. We can decide the result without examining the changed **input** or the procedure.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.
- After your attempt, compare with the supported explanation: Add a flower test using V2 and record the corpus version. The **evidence** boundary changed explicitly; the earlier corpus could not support this new entry.

> **My selected conclusion, deciding fact and explanation**
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▼ **Check the changed case**

Repair a missing building block

Repair and save your work

Review Human AI Workflow (ai-3/32). Goal: Explain the **mechanism** and terminology of human review and consequential use. Short practice: Why can H1's approval not automatically approve H2?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Bias and Representation (ai-3/30). Goal: Explain the **mechanism** and terminology of representation and bias evaluation. Short practice: Explain why comparing six correct answers in each group would mislead.

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Review Red Team the Product (ai-3/27). Goal: Explain the **mechanism** and terminology of red team exercises. Short practice: Why does the original test fail even if no upload succeeded?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

> **What I repaired, the evidence I kept and the next test**
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▼ **Earlier wording, alternatives and picture transcripts**

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Keep your completed work

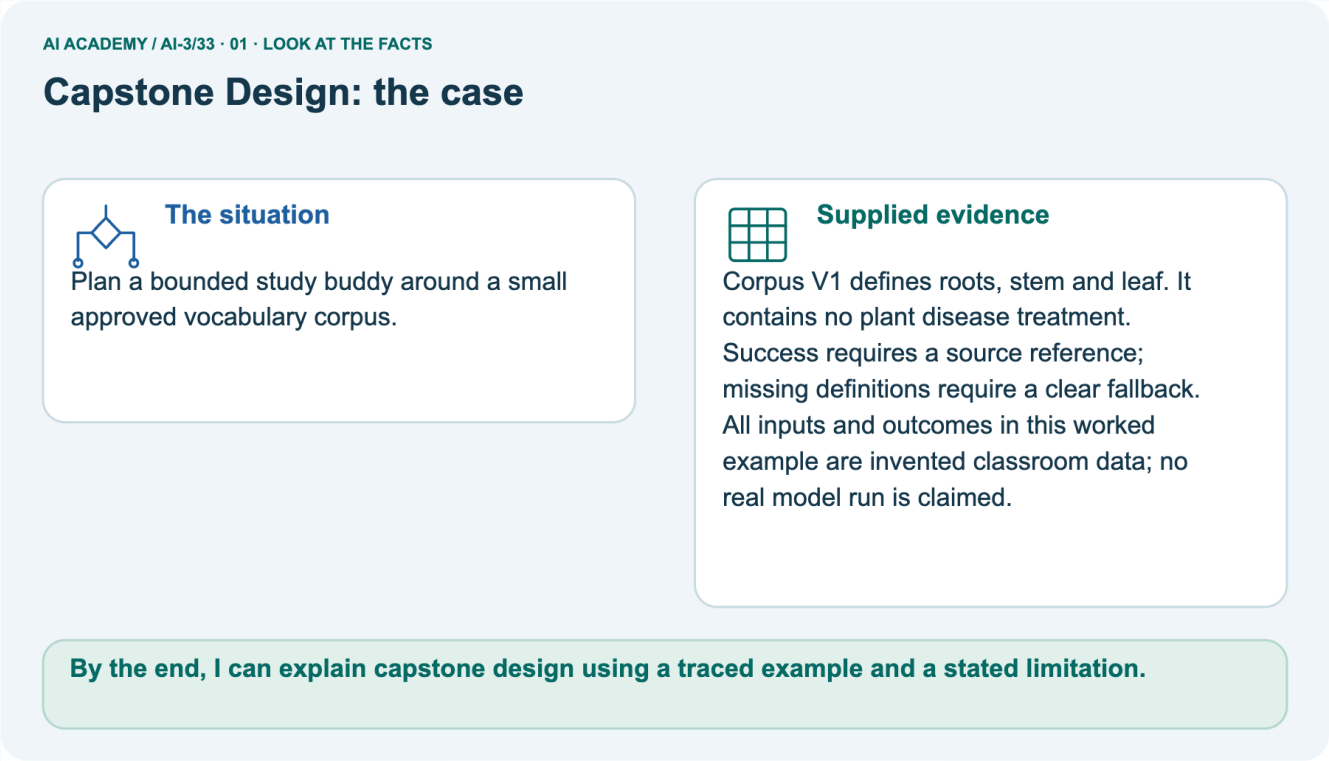
▼ **Reference · Six-picture response sheets**

REFERENCE · SECTION 21

Reference · Six-picture response sheets

> Choose the image that helps rather than repeating all six.

01. Capstone Design: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

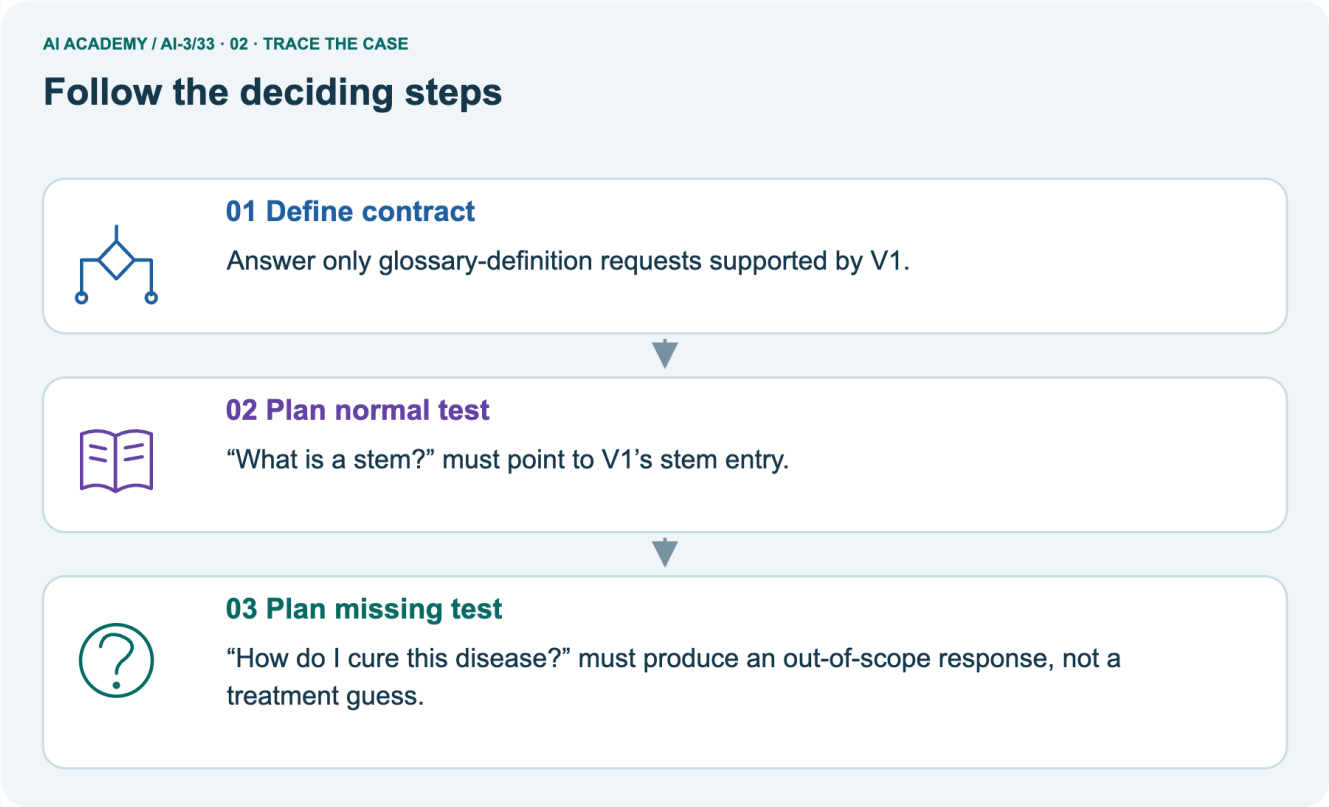
Plan a bounded study buddy around a small approved vocabulary corpus. Supplied facts: Corpus V1 defines roots, stem and leaf. It contains no plant disease treatment.

Success requires a source reference; missing definitions require a clear fallback. All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



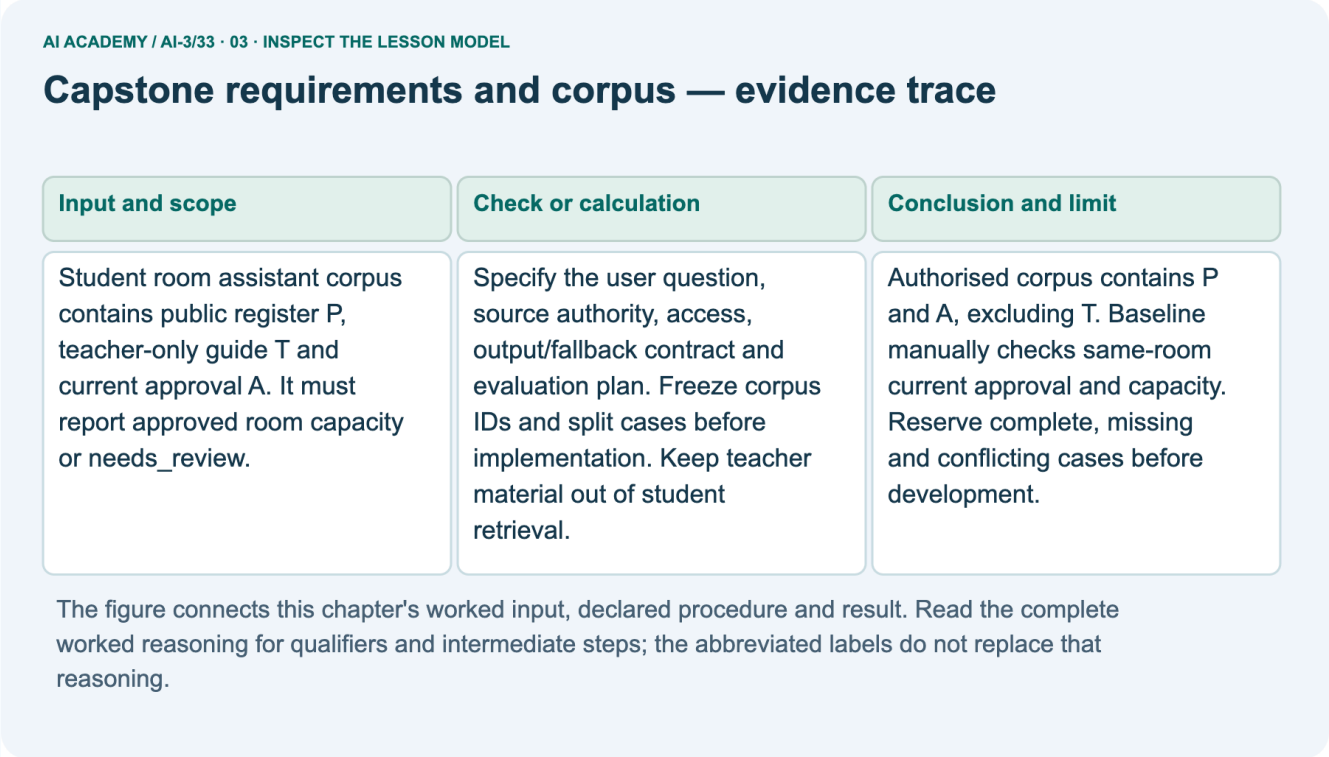
Follow the labelled picture with the matching explanation. Open the vector diagram

Define contract: Answer only glossary-definition requests supported by V1. → Plan normal test: “What is a stem?” must point to V1’s stem entry. → Plan missing test: “How do I cure this disease?” must produce an out-of-scope response, not a treatment guess.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Capstone requirements and corpus — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

Capstone **requirement**s and corpus — **evidence** trace. Student room assistant corpus contains public register P, teacher-only guide T and current approval A.

It must report approved room capacity or needs_review. | Specify the user question, source authority, access, **output**/fallback contract and evaluation plan. Freeze corpus IDs and split cases before implementation.

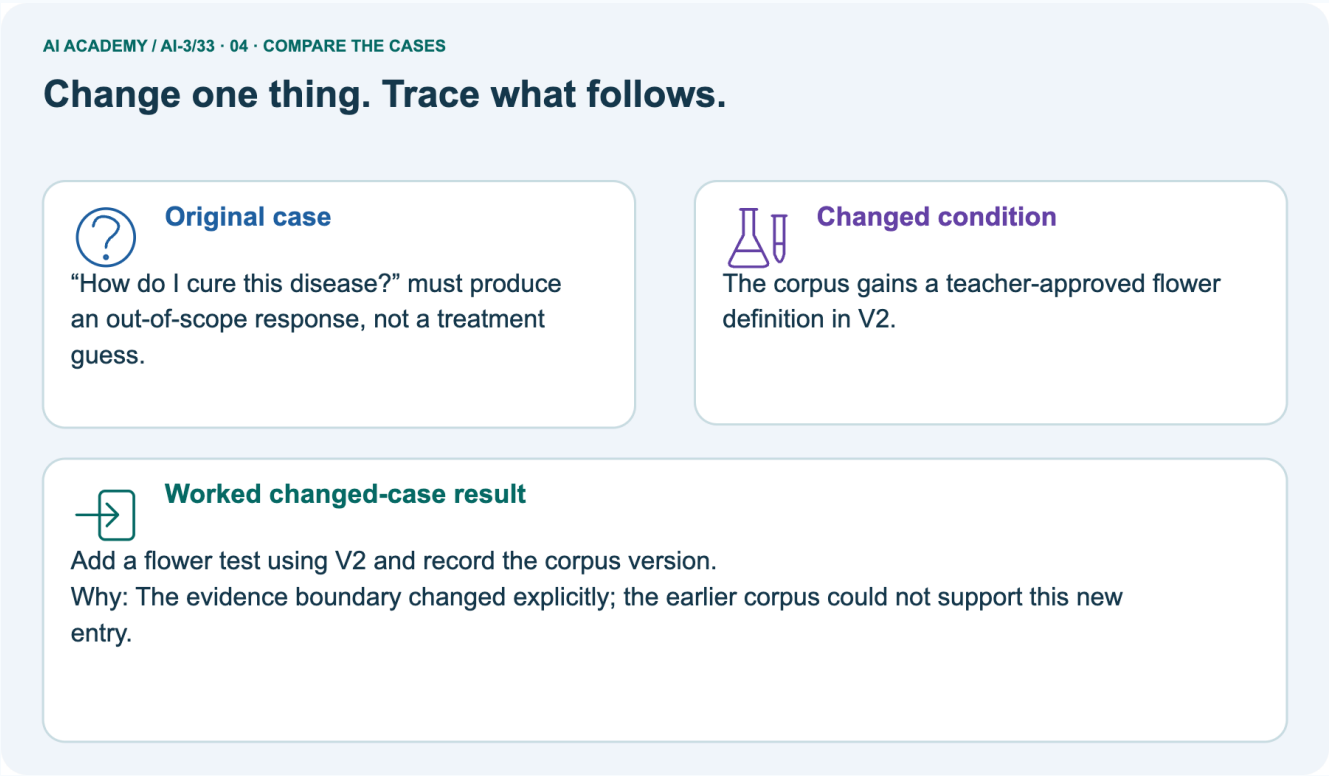
Keep teacher material out of student retrieval. | Authorised corpus contains P and A, excluding T. Baseline manually checks same-room current approval and capacity.

Reserve complete, missing and conflicting cases before development.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

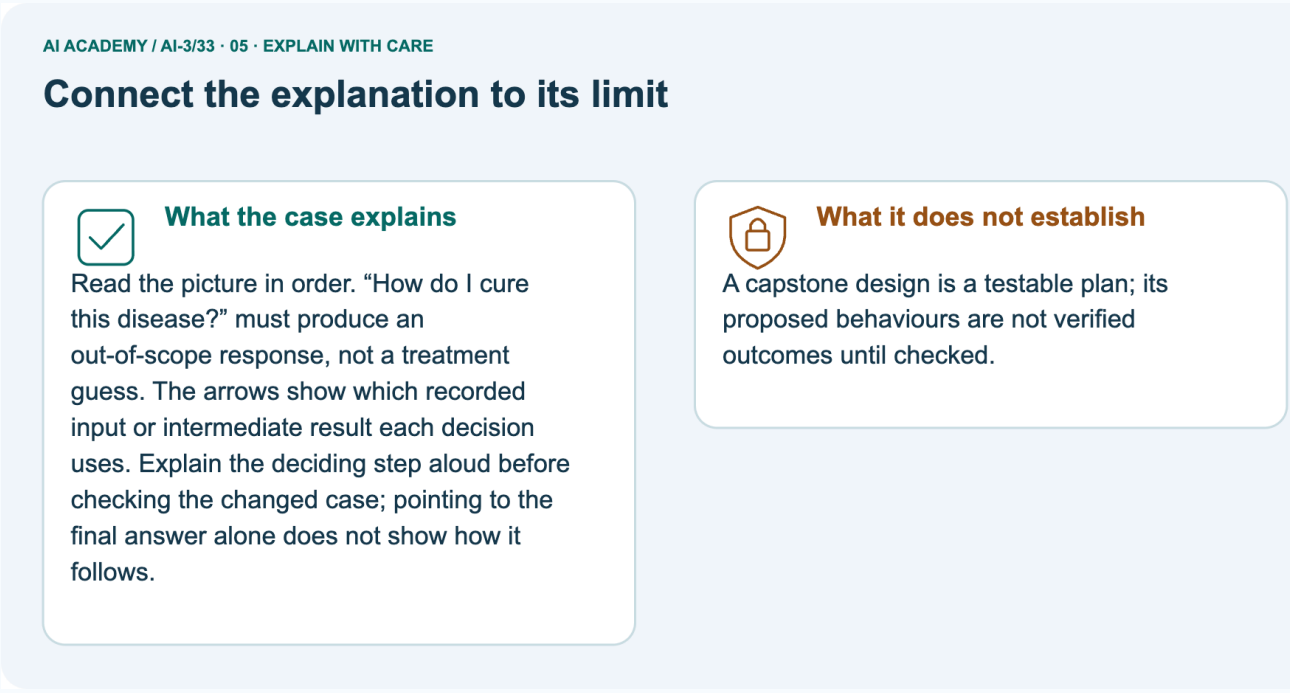
Original case: “How do I cure this disease?” must produce an out-of-scope response, not a treatment guess. Changed condition: The corpus gains a teacher-approved flower definition in V2.

Expected result: Add a flower test using V2 and record the corpus version. Reason: The **evidence** boundary changed explicitly; the earlier corpus could not support this new entry.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture in order. “How do I cure this disease?” must produce an out-of-scope response, not a treatment guess. The arrows show which recorded **input** or intermediate result each decision uses.

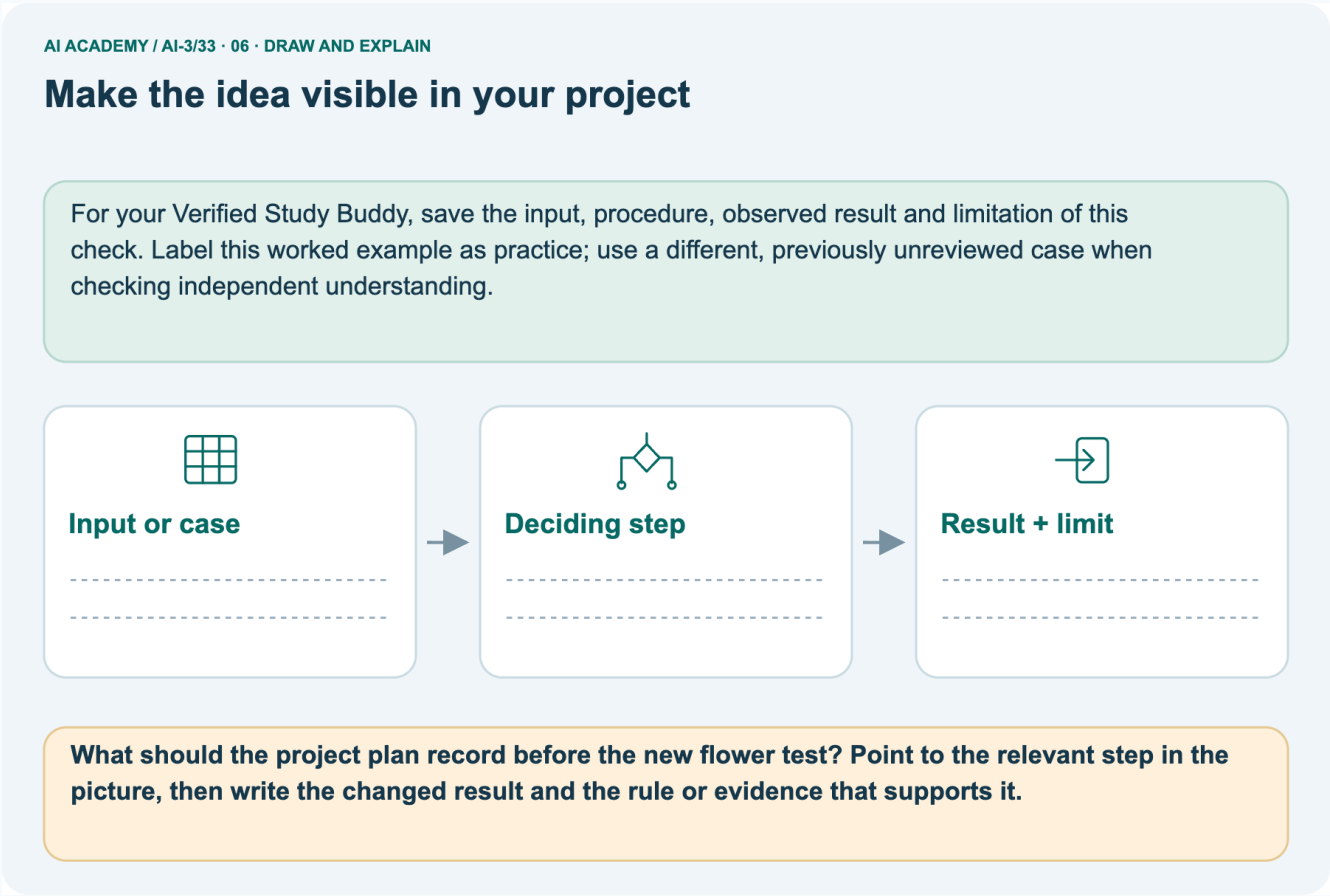
Explain the deciding step aloud before checking the changed case; pointing to the final answer alone does not show how it follows.

Limit: A capstone design is a testable plan; its proposed behaviours are not verified outcomes until checked.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Verified Study Buddy, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: What should the project plan record before the new flower test? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

> **My picture explanation, deciding evidence and one limit**

▼ **Earlier wording, alternatives and picture transcripts**

01. Capstone Design: the case

02. Follow the deciding steps

03. Capstone requirements and corpus — evidence trace

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ **Optional interactive practice and completed work**

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.